New chat

L3.pdf
PDF

Create a formal lecture scribe based off of given lecture slides. The scribe is required to serve as reference content of the exams and contain definitions, notations, assumptions, preliminaries, main results, proofs, conclusions, and worked examples as given in the lecture. Use only the lecture content. To avoid extra explanations, interpretations, and examples.

Lecturer Scribe (Structure Only)

Subject: Fundamental Ideas of Functions.

1) Definitions and Notations
Definition 1.1 (Function)

A mapping of a set A into a set B is called a function whereby every element in set A is a one to one match in set B.

Notation

Let
$f : A \to B$

where:
A = domain
B = codomain
f(x) = f (input x) = output.

2) Assumptions and Preliminaries.

The domain and codomain are non- empty sets.

Every input is associated with one output.

3) Main Result
Proof

Let $x \in A$. Definitely, there is a unique $y \in B$ that satisfies the following:

$f(x) = y$

This qualifies as a definition of a function.

Conclusion

A function is a unique element-codomain mapping.

4) Working Examples
Example 1

Let

$f(x) = x^2$

If $x = 2$, then

$f(2) = 4$

Instruction

Based on the very same format as the one above, create a lecture scribe of lecture slides.

Strict Requirements

You should observe all these rules:

Same structure and sections order:

Definitions and Notations

Assumptions and Preliminaries

Main Result

Proof

Conclusion

Working Examples

Use only the information available in the given lecture slides.

Do NOT add:

External textbook content

Extra explanations

Additional examples

Unstated assumptions.

In case some part is missing in the lecture slides, leave that part blank.

Keep academic style and tone of reference in exams.

Maintain Definitions. Definitions should be preserved as they are given. Terminology. Terminology must also be maintained as given. Logic. Logic must also be maintained as given. Examples. Examples must also be maintained as given.

Do not add questions, explanation, or reasoning to the lecture material.

Lecture Scribe – Lecture 3: Introduction to Probability Theory

Subject: CSE400 – Fundamentals of Probability in Computing

Source: Lecture Slides

1) Definitions and Notations

Definition 1.1 (Course Title)

CSE400 – Fundamentals of Probability in Computing.

Definition 1.2 (Lecture Title)

Lecture 3: Introduction to Probability Theory.

Notations

Instructor:

Dr. Dhaval Patel, PhD

Associate Professor

Machine Intelligence, Computing and xG Networks (MICxN) Research Lab

SEAS–Ahmedabad University, Ahmedabad, Gujarat–India

Course Website:

Section–1: https://campuswire.com/c/G20541104/ ↗

Section–2: https://campuswire.com/c/G766840CE/ ↗

Lecture Sessions:

Section–1:

9:30 AM to 11:00 AM (Tuesday, Thursday) – GICT Room–136

Section–2:

1:00 PM to 2:30 PM (Tuesday, Thursday) – GICT Room–137

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2) Assumptions and Preliminaries

The course includes the following components:

General Course Information

- Team

- Active Learning Platform: Campuswire
- Schedule
- Grading

Active Learning and Class Discussion

Course Website Sections:

- Section–1: Campuswire
- Section–2: Campuswire

Campuswire Usage

Campuswire is used for:

- To build confidence level: Participate anonymously, post/back-channel communication during lecture
- Collaborative and active learning: Solve difficulties with online participation
- Feedback on lecture session: Real-time feedback and polling
- Personal communication with instructor/TAs: Direct message communication

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Instructor Interaction

Timing for Discussion and Difficulty Sessions:

- Contact Hours: 24 × 7 through Campuswire
- Best Practice: Put queries on Campuswire
- Need more time: Direct message communication between student and instructor
- External Engagement: UGRP–8: 2026 / Join with offline projects / Need counseling / Cup of tea for 10 minutes / Write on office email ID

Important Note

LaTeX tutorial and assignment submission guideline.

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3) Main Result

Proof

The lecture introduces the course structure, instructor information, course platform, lecture sessions, and project execution guidelines.

The project execution consists of:

Major Milestones (M1–M6):

M1: Kickstart:
Team formation, area identification, background, motivation, and problem formulation

M2: Mathematical Modeling:
Mathematical modeling of the selected problem domain

M3: Coding:
Simulation and computation

M4: Inference:
Choose, understand, and code a randomized algorithm

M5: Randomized Algorithms:
Apply randomized algorithm to the domain problem and obtain results

M6: Derive Bounds, Analysis, and Final Deliverables Submission

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Conclusion

The lecture establishes the course structure, communication platform, instructor interaction process, lecture schedule, and project execution milestones for CSE400.

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4) Working Examples

Example 1 (Engineering Applications)

Engineering Applications include:

- Speech Recognition
- System Radar System
- Communication Network

Example 2 (Project Deliverables)

Project Deliverables include:

- Codes
- Reports
- Videos
- Multimodal artifacts
- Decision logs
- Project demonstration videos

End of Scribe